



Post Office Box 2000, Decatur, Alabama 35609-2000

October 29, 2024

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Units 1 and 2
Renewed Facility Operating License Nos. DPR-33 and DPR-52
NRC Docket Nos. 50-259 and 50-260

Subject: Licensee Event Report 50-259/2024-003-00

The enclosed Licensee Event Report provides details of a breaker trip which automatically started Emergency Diesel Generators on Browns Ferry Nuclear Plant, Units 1 and 2. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(iv)(A), as an event or condition that resulted in the unplanned, valid, manual or automatic actuation of emergency AC electrical power systems.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact David J. Renn, Nuclear Site Licensing Manager, at (256) 729-2636.

Respectfully,

A handwritten signature in black ink, appearing to read "DLAK", is written over a horizontal line.

Daniel A. Komm
Site Vice President

Enclosure: Licensee Event Report 50-259/2024-003-00 – Valid Specified System
Actuation Caused the Automatic Start of Emergency Diesel Generators

cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant
NRC Project Manager - Browns Ferry Nuclear Plant

U.S. Nuclear Regulatory Commission
Page 2
October 29, 2024

DJR:RWC
Enclosure
bcc (w/ Enclosure):

A. Aboulfaida
K. S. Adams
M. B. Bruce
D. M. Delk
R. C. Dreke
S. T. Earley
C. Edmondson
D. E. Ferrell
J. E. Gordon
K. D. Hulvey
J. T. Johnson
D. Komm
E. Q. Leonard
R. Medina
M. W. Oliver
J. L. Paul
M. Rasmussen
T. S. Rausch
D. J. Renn
C. L. Rice
D. K. Riggs
M. Sivaraman
J. A. Yarbrough
ECM

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 04/30/2027				
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nureqs/staff/sr1022/r3/)				Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: pira_submission@omb.eop.gov . The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.				
1. Facility Name Browns Ferry Nuclear Plant, Unit 1				☒ 050 ☐ 052		52. Docket Number 00259				
3. Page 1 OF 6										
4. Title Valid Specified System Actuation Caused the Automatic Start of Emergency Diesel Generators										
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
08	30	2024	2024	- 003 -	00	10	29	2024	Browns Ferry Nuclear Plant, Unit 2	05000260
									Facility Name	Docket Number
									N/A	05000XXX
9. Operating Mode 1						10. Power Level 73				
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)										
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)		☐ 21.21(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)				☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)				
☐ OTHER (Specify here, in abstract, or NRC 366A).										
12. Licensee Contact for this LER										
Licensee Contact Ryan Coons, Licensing Engineer								Phone Number (Include area code) 256-729-2070		
13. Complete One Line for each Component Failure Described in this Report										
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS	
X	EK	BKR	G080	Y	N/A	N/A	N/A	N/A	N/A	N/A
4. Supplemental Report Expected)					15. Expected Submission Date			Month	Day	Year
☒ No ☐ Yes (If yes, complete 15. Expected Submission Date								N/A	N/A	N/A
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)										
On August 30, 2024, at 1051 Central Daylight Time Breaker 1622 failed to close during the transfer of Shutdown Bus 1 from 4kV Unit Board 1A to 4kV Unit Board 2B. This failure resulted in deenergizing 4kV Shutdown (SD) Boards A and B, and the A and B Emergency Diesel Generators started and tied to their respective 4kV SD Boards in response. The cause of the event was a loose LS-9 appliance lug which resulted in a loss of power on 4kV SD Boards A and B. As corrective action, the lug was repaired. Additional actions to prevent recurrence will include, briefing of operators and maintenance personnel on this event, as well as procedure enhancements to operating instructions.										

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 1	☒ 050	2. DOCKET NUMBER 00259	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 003	REV NO. - 00

NARRATIVE**I. Plant Operating Conditions before the Event**

At the time of occurrence, Browns Ferry Nuclear Plant (BFN) Unit 1 was in Mode 1 at approximately 73 percent power. Power decreased throughout this event as part of a planned shutdown to support Unit 1 Refueling Outage 15 (U1R15). BFN Unit 2 was in Mode 1 at approximately 100 percent power.

II. Description of Event**A. Event Summary**

On August 30, 2024, at 1051 Central Daylight Time (CDT) Breaker 1622 (BFN-0-BKR-211-000D/001) [BKR] failed to close during the planned transfer of Shutdown Bus 1 from 4kV Unit Board 1A to 4kV Unit Board 2B in preparation for U1R15. This failure resulted in deenergizing 4kV Shutdown Boards (SD BDs) A and B, and the A and B Emergency Diesel Generators (EDGs) [EK] automatically started and tied to their respective 4kV SD BDs in response.

The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(iv)(A), as an event or condition that resulted in the manual or automatic actuation of emergency alternating current (AC) electrical power systems, including: EDGs.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, systems, or components (SSCs) whose inoperability contributed to this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 1	☒ 050	2. DOCKET NUMBER 00259	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 003	REV NO. - 00

NARRATIVE**C. Dates and approximate times of occurrences**

<u>DATES AND APPROXIMATE TIMES</u>	<u>OCCURRENCE</u>
August 30, 2024 1051 CDT	4kV Shutdown Bus 1 failed to transfer. Breaker 1622 failed to close, and BFN, Unit 1, lost power to the A and B 4kV SD BDs. EDGs A and B started and tied to their respective boards.
August 30, 2024 1250 CDT	4kV SD BD A energized through the normal feeder breaker.
August 30, 2024 1336 CDT	4kV SD BD B energized through the normal feeder breaker.
August 30, 2024 1730 CDT	Ops personnel reported this event to the Nuclear Regulatory Commission (NRC) as Event Notification 57298, in accordance with 10 CFR 50.72(b)(3)(iv)(A).

D. Manufacturer and model number of each component that failed during the event

BFN-0-BKR-211-000D/001 (Breaker 1622) is a General Electric circuit breaker, Part Number 317A7502P005.

E. Other systems or secondary functions affected

The equipment powered by the SD BD lost power until restarted following re-energization of the board.

F. Method of discovery of each component or system failure or procedural error

The failure of the breaker was discovered to be a loose lug that was discovered during troubleshooting of the breaker.

G. The failure mode, mechanism, and effect of each failed component

The breaker failed due to a loose LS-9 appliance lug.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 1	☒ 050	2. DOCKET NUMBER 00259	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 003	REV NO. - 00

NARRATIVE**H. Operator actions**

The operators ensured the EDGs started and supplied power to the shutdown board and shutdown the EDGs once the normal power supply was restored to the SD boards.

I. Automatically and manually initiated safety system responses

The loss of power to the SD BDs is condition also caused a loss of the 1A and 2A reactor protection system (RPS), resulting in the invalid actuation of Primary Containment Isolation System Groups 2, 3, and 6 on BFN, Units 1 and 2.

The loss of the "A" RPS on BFN, Units 1 and 2 was not a specified system actuation, only the actuation of the "A" and "B" EDGs were a specified system actuation.

III. Cause of the event

The direct cause of the event was a loose LS-9 appliance lug which resulted in a loss of power on 4kV SD BDs A and B. There had been previous issues with this appliance lug coming loose which had yet to be permanently corrected.

A. Cause of each component or system failure or personnel error

The LS-9 appliance lug is uniquely specific to this application. The LS-9 appliance lug inputs into the charging spring indicating light. The LS-9 appliance lug required replacement, however, the required tooling had not been permitted for use at TVA.

B. Cause(s) and circumstances for each human performance related root cause

No human performance related root causes were identified.

IV. Analysis of the event

The 4kV bus under-voltage condition was a valid EDG signal, which caused the EDG to fulfil its designed safety function by automatically starting and tying to the board.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 1	☒ 050	2. DOCKET NUMBER 00259	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 003	REV NO. - 00

NARRATIVE**V. Assessment of Safety Consequences**

All safety systems and alarms performed as designed in response to the unplanned, valid, automatic actuation signal. No actual safety consequences or radiological releases resulted from this event. All Technical Specification and Technical Requirements Manual conditions were met throughout this event. All mitigating systems remained functional and capable of fulfilling their required safety functions throughout this event. No fission product barriers were challenged during this event.

Based on the above, the TVA has concluded that sufficient systems were available to provide the required safety functions needed to protect the health and safety of the public.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

The breaker's failure caused a valid actuation signal for the EDGs, which continued to power the affected SD BDs.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shut down.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

No safety systems were rendered inoperable during this event.

VI. Corrective Actions

Corrective Actions are being managed by the TVA's Corrective Action Program (CAP) under Condition Report (CR) 1954797.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 1	☒ 050	2. DOCKET NUMBER 00259	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 003	REV NO. - 00

NARRATIVE**A. Immediate Corrective Actions**

Troubleshooting on Breaker 1622 and identified a loose wire on the charging spring limit switch. This wire was tightened and reattached.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

To prevent a recurrence of this event, BFN will perform the following corrective actions:

- Verify the completion of Work Order (WO) 124788330, which replaces the LS-9 appliance lug for BFN-0-BKR-211-000D/001.
- Verify that WOs exist to inspect safety-related 4kV Siemens breakers identified in the extent of condition and replaces their appliance lugs as required.
- Conduct briefings to operations and maintenance personnel regarding this event.
- Revise 0-OI-57A, Switchyard and 4160V AC Electrical System, to include the guidance on precautions and limitations associated with the charging spring failure light.

VII. Previous Similar Events at the Same Site

A review of the BFN Licensee Event Reports (LERs) over the last five years revealed two similar events at BFN:

- LER 50-296/2024-002-00 describes a breaker trip which automatically started an EDG on BFN, Unit 3. This event was believed to have been caused by an intermittent loose connection, which was resolved during troubleshooting activities.
- LER 50-296/2020-001-00 describes an offsite lightning strike which caused the loss of the 4 kV SD BDs on BFN, Unit 3, and automatic actuated its EDGs.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.